

JYOTHI SUMITRA VADDI

AI Engineer | Generative AI | Python | RAG

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PROFESSIONAL SUMMARY

Computer Science undergraduate with hands-on experience building **Generative AI applications, Retrieval-Augmented Generation (RAG) systems, and AI-powered APIs** using Python. Experienced in working with **LangChain, LangGraph, LLMs, embeddings, vector databases, FastAPI, and ChromaDB**. Built project-based AI applications involving document processing, semantic search, prompt engineering, and conversational interfaces. Strong foundation in Python, data structures, problem-solving, and software development. Currently developing practical experience in cloud deployment, Docker, model evaluation, and production AI systems.

TECHNICAL SKILLS

Programming: Python, SQL

AI / Machine Learning: Machine Learning, Generative AI, Large Language Models (LLMs), Prompt Engineering, RAG, AI Agents, Multi-Agent Systems

Frameworks: LangChain, LangGraph, FastAPI

Data & Retrieval: Embeddings, Vector Databases, ChromaDB, Document Processing, Semantic Search

APIs & Development: REST APIs, AI/LLM APIs, JSON, Git, GitHub

Cloud & DevOps: AWS, Docker

PROJECTS

AI PDF Assistant

Python | LangChain | LangGraph | RAG | ChromaDB | FastAPI | Gemini

- Developed a Retrieval-Augmented Generation (RAG) application that allows users to interact with uploaded PDF documents using natural language.
- Implemented document processing and text chunking pipelines for extracting meaningful information from PDF files.
- Generated embeddings and stored document representations in ChromaDB for semantic retrieval.
- Built a conversational question-answering system using an LLM and retrieved document context.
- Developed a FastAPI backend with REST endpoints for PDF uploading and conversational queries.
- Implemented document summarization and contextual question answering.
- Added intelligent routing between PDF-based questions, web search, and general conversation.
- Used LangChain and LangGraph to structure AI workflows.

AI Support Assistant

Python | LangChain | LangGraph | Gemini | FastAPI

- Built an AI support assistant capable of answering user queries using an LLM.
- Designed structured AI workflows using LangGraph nodes and conditional routing.
- Implemented prompt engineering techniques to improve response consistency.
- Integrated external AI APIs for generating responses.
- Designed REST API endpoints using FastAPI.
- Maintained conversation history to provide contextual responses.

Multi-Agent Research Assistant

Python | LangGraph | LLMs | AI Agents

- Developed a prototype multi-agent system where specialized AI agents collaborate to answer research questions.
- Designed separate agents for research, analysis, and response generation.
- Implemented conditional routing between agents using LangGraph.
- Experimented with prompt engineering and structured LLM outputs.

Student Analytics Dashboard

Python | Pandas | Data Analysis

- Developed a Python-based student analytics application for managing and analyzing student records.
- Implemented data searching, grading, statistics, and performance analysis.
- Used Pandas for structured data processing and analysis.

EXPERIENCE

AI / Software Development Intern

QSkill | Remote | June 2026 – July 2026

- Developed frontend applications using React and Tailwind CSS.
- Worked on task-based software development assignments.
- Implemented reusable UI components and application logic.
- Practiced debugging, version control, and collaborative development workflows.

CERTIFICATIONS

- AWS Certified Cloud Practitioner
- Generative AI Foundations
- Introduction to Large Language Models
- Python for Data Science and AI
- Foundation: Introduction to LangChain

EDUCATION

Bachelor of Technology in Computer Science and Engineering

Koneru Lakshmaiah Education Foundation | 2024 – 2028

Relevant coursework: Data Structures and Algorithms, Linear Algebra, Python Programming, Database Management Systems, Artificial Intelligence, Machine Learning, Computer Networks

ADDITIONAL INFORMATION

- Strong interest in Generative AI and AI Engineering
- Interested in RAG, AI agents, and LLM application development
- Strong problem-solving and analytical skills
- Comfortable learning new AI frameworks and technologies
- Interested in building production-oriented AI applications